Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load Dataset

```
df = pd.read_csv('/content/mymoviedb.csv', lineterminator='\n')
df.head()
```

| | Release_Date | Title | Overview | Popularity | Vote_Count | Vote_Average | Original_Language | Genre | Poster_Url |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 2021-12-15 | Spider-Man: No Way Home | Peter Parker is unmasked and no longer able to... | 5083.954 | 8940 | 8.3 | en | Action, Adventure, Science Fiction | https://image.tmdb.org/t/p/original/1g0dhYtq4i... |
| **1** | 2022-03-01 | The Batman | In his second year of fighting crime, Batman u... | 3827.658 | 1151 | 8.1 | en | Crime, Mystery, Thriller | https://image.tmdb.org/t/p/original/74xTEgt7R3... |
| **2** | 2022-02-25 | No Exit | Stranded at a rest stop in the mountains durin... | 2618.087 | 122 | 6.3 | en | Thriller | https://image.tmdb.org/t/p/original/vDHsLnOWKl... |
| **3** | 2021-11-24 | Encanto | The tale of an extraordinary family, the Madri... | 2402.201 | 5076 | 7.7 | en | Animation, Comedy, Family, Fantasy | https://image.tmdb.org/t/p/original/4j0PNHkMr5... |
| **4** | 2021-12-22 | The King's Man | As a collection of history's worst tyrants and... | 1895.511 | 1793 | 7.0 | en | Action, Adventure, Thriller, War | https://image.tmdb.org/t/p/original/aq4Pwv5Xeu... |

Next steps: Generate code with df  New interactive sheet

Basic Exploration

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Release_Date       9827 non-null   object
 1   Title              9827 non-null   object
```

```
 2   Overview           9827 non-null   object
 3   Popularity         9827 non-null   float64
 4   Vote_Count         9827 non-null   int64
 5   Vote_Average       9827 non-null   float64
 6   Original_Language  9827 non-null   object
 7   Genre              9827 non-null   object
 8   Poster_Url         9827 non-null   object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB
```

```
df.describe()
```

| | Popularity | Vote_Count | Vote_Average |
|---|---|---|---|
| **count** | 9827.000000 | 9827.000000 | 9827.000000 |
| **mean** | 40.326088 | 1392.805536 | 6.439534 |
| **std** | 108.873998 | 2611.206907 | 1.129759 |
| **min** | 13.354000 | 0.000000 | 0.000000 |
| **25%** | 16.128500 | 146.000000 | 5.900000 |
| **50%** | 21.199000 | 444.000000 | 6.500000 |
| **75%** | 35.191500 | 1376.000000 | 7.100000 |
| **max** | 5083.954000 | 31077.000000 | 10.000000 |

```
df['Genre'].head()
```

| | Genre |
|---|---|
| **0** | Action, Adventure, Science Fiction |
| **1** | Crime, Mystery, Thriller |
| **2** | Thriller |
| **3** | Animation, Comedy, Family, Fantasy |
| **4** | Action, Adventure, Thriller, War |

**dtype:** object

```
df.duplicated().sum()
```

```
np.int64(0)
```

Convert Date column

```
df['Release_Date'] = pd.to_datetime(df['Release_Date'])
```

```
df['Release_Date'] = df['Release_Date'].dt.year
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Release_Date       9827 non-null   int32
 1   Title              9827 non-null   object
 2   Overview           9827 non-null   object
 3   Popularity         9827 non-null   float64
 4   Vote_Count         9827 non-null   int64
 5   Vote_Average       9827 non-null   float64
 6   Original_Language  9827 non-null   object
 7   Genre              9827 non-null   object
 8   Poster_Url         9827 non-null   object
dtypes: float64(2), int32(1), int64(1), object(5)
memory usage: 652.7+ KB
```

Drop Uncessary columns

```
cols = ['Overview', 'Original_Language', 'Poster_Url']
df.drop(cols, axis=1, inplace=True)
```

```
df.head()
```

| | Release_Date | Title | Popularity | Vote_Count | Vote_Average | Genre |
|---|---|---|---|---|---|---|
| **0** | 2021 | Spider-Man: No Way Home | 5083.954 | 8940 | 8.3 | Action, Adventure, Science Fiction |
| **1** | 2022 | The Batman | 3827.658 | 1151 | 8.1 | Crime, Mystery, Thriller |
| **2** | 2022 | No Exit | 2618.087 | 122 | 6.3 | Thriller |
| **3** | 2021 | Encanto | 2402.201 | 5076 | 7.7 | Animation, Comedy, Family, Fantasy |
| **4** | 2021 | The King's Man | 1895.511 | 1793 | 7.0 | Action, Adventure, Thriller, War |

Next steps: Generate code with df  New interactive sheet

Categorize vote average

```
def categorize_col(df, col, labels):
    """
    Categorizes a column based on quartiles
    """
    edges = [
        df[col].describe()['min'],
        df[col].describe()['25%'],
        df[col].describe()['50%'],
        df[col].describe()['75%'],
        df[col].describe()['max']
    ]

    df[col] = pd.cut(df[col], edges, labels=labels, duplicates='drop')
    return df
```

Apply Function

```
labels = ['not_popular', 'below_avg', 'average', 'popular']
df = categorize_col(df, 'Vote_Average', labels)
```

```
df['Vote_Average'].value_counts()
```

| Vote_Average | count |
|---|---|
| **not_popular** | 2467 |
| **popular** | 2450 |
| **average** | 2412 |
| **below_avg** | 2398 |

**dtype:** int64

Handle Missing Values

```
df.dropna(inplace=True)
```

```
df.isna().sum()
```

| | 0 |
|---|---|
| **Release_Date** | 0 |
| **Title** | 0 |
| **Popularity** | 0 |
| **Vote_Count** | 0 |
| **Vote_Average** | 0 |
| **Genre** | 0 |

**dtype:** int64

Split and Explode Genre

```
df['Genre'] = df['Genre'].str.split(', ')
```

```
df = df.explode('Genre').reset_index(drop=True)
```

```
df.head()
```

| | Release_Date | Title | Popularity | Vote_Count | Vote_Average | Genre |
|---|---|---|---|---|---|---|
| **0** | 2021 | Spider-Man: No Way Home | 5083.954 | 8940 | popular | Action |
| **1** | 2021 | Spider-Man: No Way Home | 5083.954 | 8940 | popular | Adventure |
| **2** | 2021 | Spider-Man: No Way Home | 5083.954 | 8940 | popular | Science Fiction |
| **3** | 2022 | The Batman | 3827.658 | 1151 | popular | Crime |
| **4** | 2022 | The Batman | 3827.658 | 1151 | popular | Mystery |

Next steps: Generate code with df  New interactive sheet

Convert Genre to Category

```
df['Genre'] = df['Genre'].astype('category')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
```

```
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Release_Date  25552 non-null  int32
 1   Title         25552 non-null  object
 2   Popularity    25552 non-null  float64
 3   Vote_Count    25552 non-null  int64
 4   Vote_Average  25552 non-null  category
 5   Genre         25552 non-null  category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 749.6+ KB
```

```
df.nunique()
```

| | 0 |
|---|---|
| **Release_Date** | 100 |
| **Title** | 9415 |
| **Popularity** | 8088 |
| **Vote_Count** | 3265 |
| **Vote_Average** | 4 |
| **Genre** | 19 |

**dtype:** int64

Visualization Setup

```
sns.set_style('whitegrid')
```

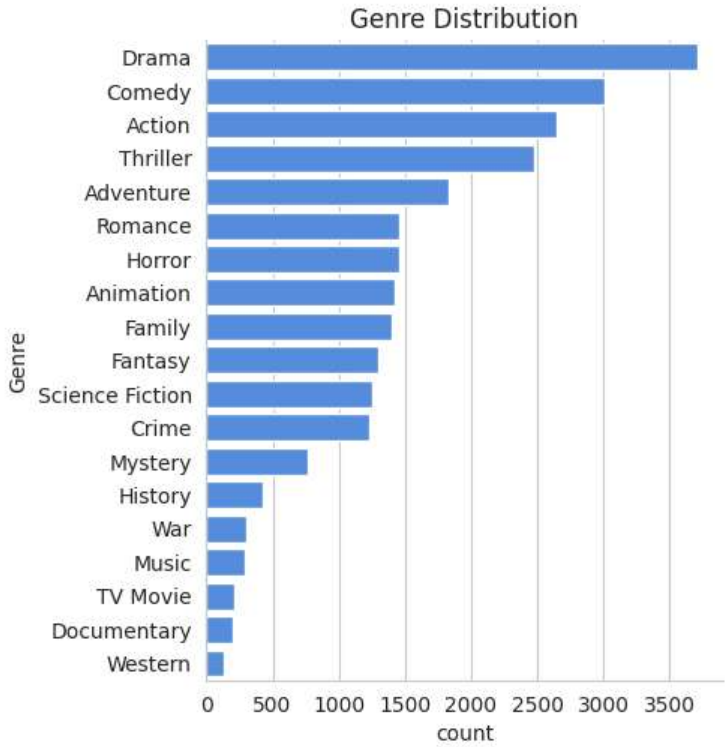
Q1: Most Frequent Genre

```
df['Genre'].describe()
```

| | Genre |
|---|---|
| **count** | 25552 |
| **unique** | 19 |
| **top** | Drama |
| **freq** | 3715 |

**dtype:** object

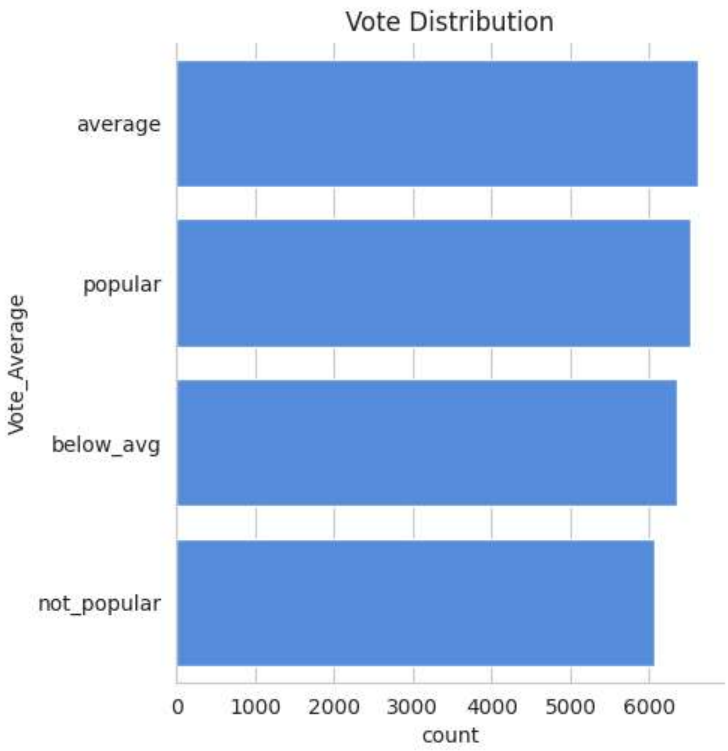
```
sns.catplot(
    y='Genre',
    data=df,
    kind='count',
    order=df['Genre'].value_counts().index,
    color='#4287f5'
)

plt.title('Genre Distribution')
plt.show()
```




Q2: Vote Distribution

```
sns.catplot(
    y='Vote_Average',
    data=df,
    kind='count',
    order=df['Vote_Average'].value_counts().index,
    color='#4287f5'
)
```

```
plt.title('Vote Distribution')
plt.show()
```




Highest Popularity Movie

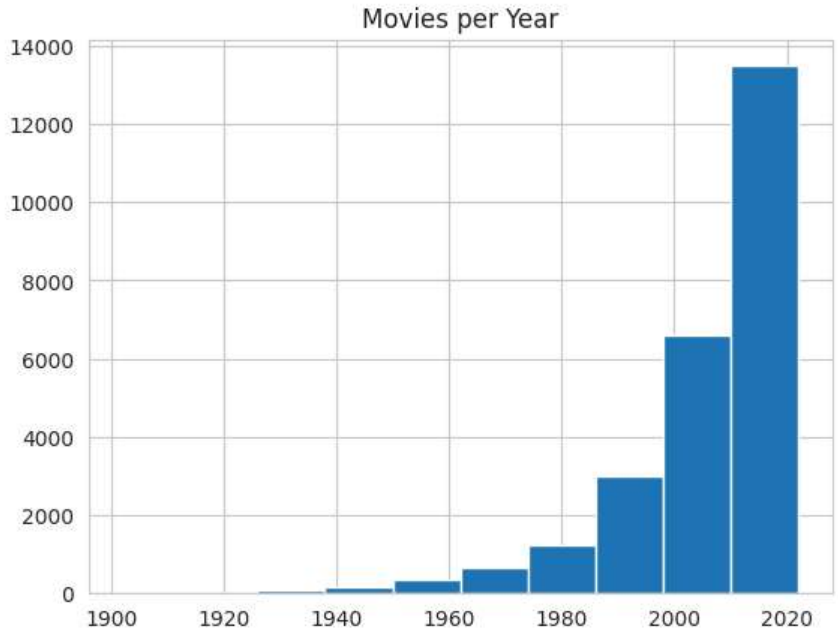
```
df[df['Popularity'] == df['Popularity'].max()]
```

| | Release_Date | Title | Popularity | Vote_Count | Vote_Average | Genre |
|---|---|---|---|---|---|---|
| **0** | 2021 | Spider-Man: No Way Home | 5083.954 | 8940 | popular | Action |
| **1** | 2021 | Spider-Man: No Way Home | 5083.954 | 8940 | popular | Adventure |
| **2** | 2021 | Spider-Man: No Way Home | 5083.954 | 8940 | popular | Science Fiction |

Q4: Lowest Popularity Movie

```
df[df['Popularity'] == df['Popularity'].min()]
```

| | Release_Date | Title | Popularity | Vote_Count | Vote_Average | Genre |
|---|---|---|---|---|---|---|
| **25546** | 2021 | The United States vs. Billie Holiday | 13.354 | 152 | average | Music |
| **25547** | 2021 | The United States vs. Billie Holiday | 13.354 | 152 | average | Drama |
| **25548** | 2021 | The United States vs. Billie Holiday | 13.354 | 152 | average | History |
| **25549** | 1984 | Threads | 13.354 | 186 | popular | War |
| **25550** | 1984 | Threads | 13.354 | 186 | popular | Drama |
| **25551** | 1984 | Threads | 13.354 | 186 | popular | Science Fiction |

Q5: Movies per Year

```
df['Release_Date'].hist()
plt.title('Movies per Year')
plt.show()
```